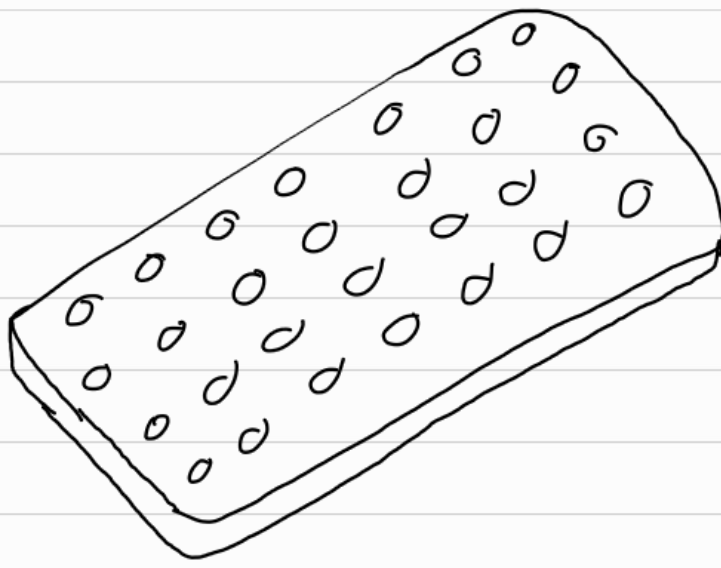
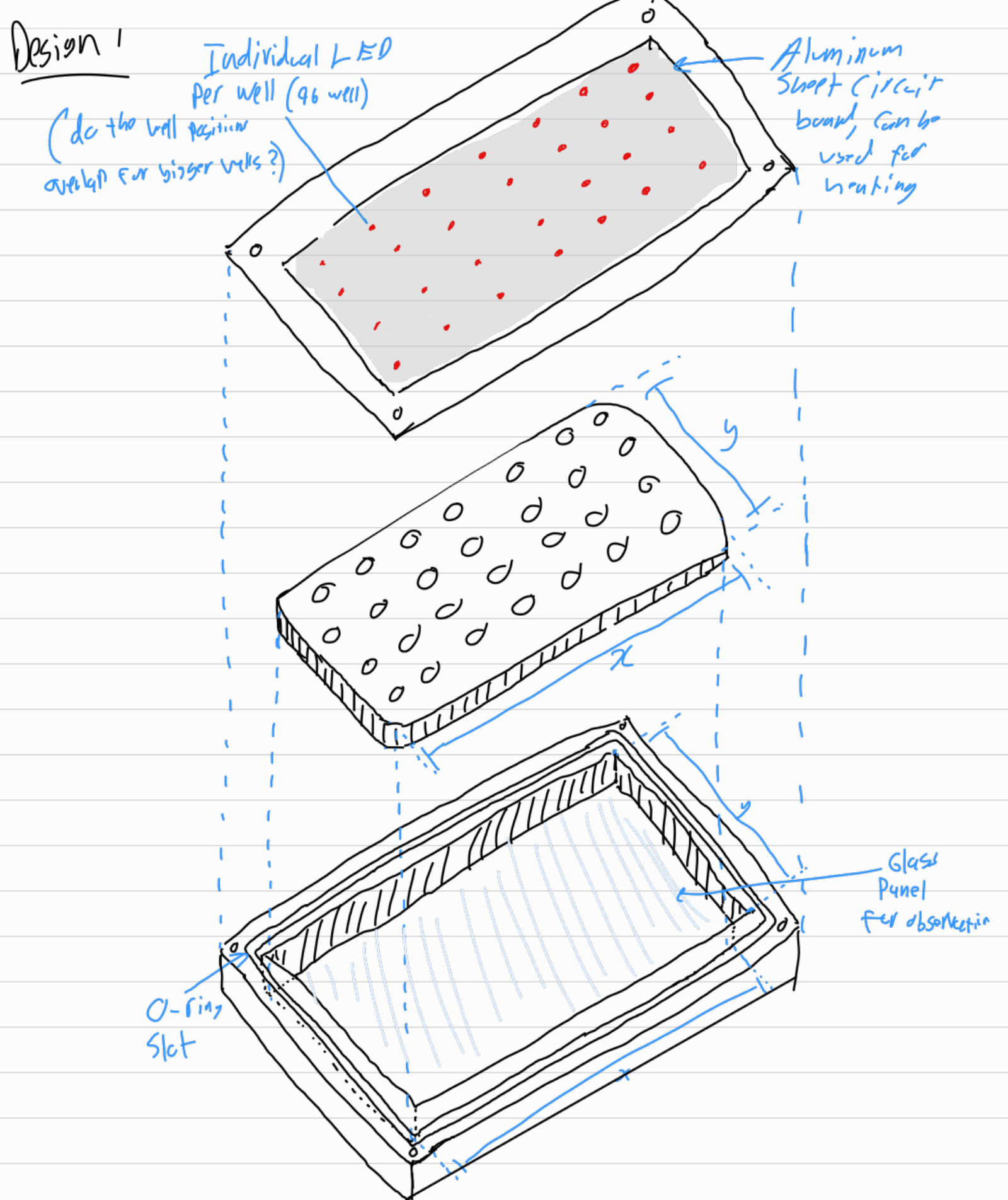






- The top plate is a fully Aluminium with 1 LED per well (for 96-well) (Plate is $135 \times 92 \text{ mm} \times 1.1 \text{ mm}$)

- This allows for heating to be done through both the plate as well as the top aluminium lid

- The Aluminium plate acts as a heat sink/distribution plate

- Solid opaque top lid, can't see the wells from the top
- Using 1 led per well for 96-well plate might restrict compatibility with different number of well plates

- No need for glass top, which may be complex/costly

- Aluminium plate mounts to a lid chassis. This then mounts to the base plate with an O-ring in between for sealing purposes

1 Silicone rubber gasket $\rightarrow$ $\begin{matrix} \text{ID} & 46.3 \text{ mm} \\ \text{OD} & 4 \text{ mm} \\ \text{diameter} \end{matrix}$

- Use 4 screws to mount the top and bottom plate (M4 Bolts, 27mm)

- I'd really want a min of IP68 rating for sensors (dust tight + continuous liquid immersion)

- Bottom plate and lid has a groove for the gasket

- Base has Hexagon holes to fit Nuts for bolts

- Base has a Slot for a Glass pane on the back (2.1mm thick)
- Designed to Fit an Ibidi N-plate 96 Well with dimensions 127.8 x 85.5 x 16.6 mm with the lid on.

Lid System:

- Can we use an aluminium top cover?
 - ↳ No Visibility of culture media from the top
 - ↳ Lid will have built in LED Grid to illuminate the wells
 - ↳ Lid will provide better heat distribution
 - ↳ No need to worry about Condensation
 - ↳ Can LEDs handle conditions in an autoclave?
- If want glass top, how important is the optical properties of the glass?
 - ↳ Heated glass products such as ITO coated glass which is a glass pane coated in a transparent conductive material that can be used to heat the entire glass pane however they are often quite pricey
 - e.g. from shilpent.com 10 pack of 100 x 100 x 1.1 mm with 10 ohm/sq costs ~ \$155 ~ £2478.14
 - ↳ otherwise there may be a way to heat normal glass with less control and possibly more optically obstructive using thin heating elements bonded to the top of the glass

↳ otherwise we could heat the glass from the sides using traditional heating elements, but this would reduce temperature control and distribution

Another option recommended by Justin:



- Channel between two gaskets used as a barrier + fail safe.
IF Inner gasket fails $\Rightarrow$ potential hazards inside are exposed and
Sterilized + indication of fail
- IF outer gasket fails $\Rightarrow$ No contamination + indication of fail

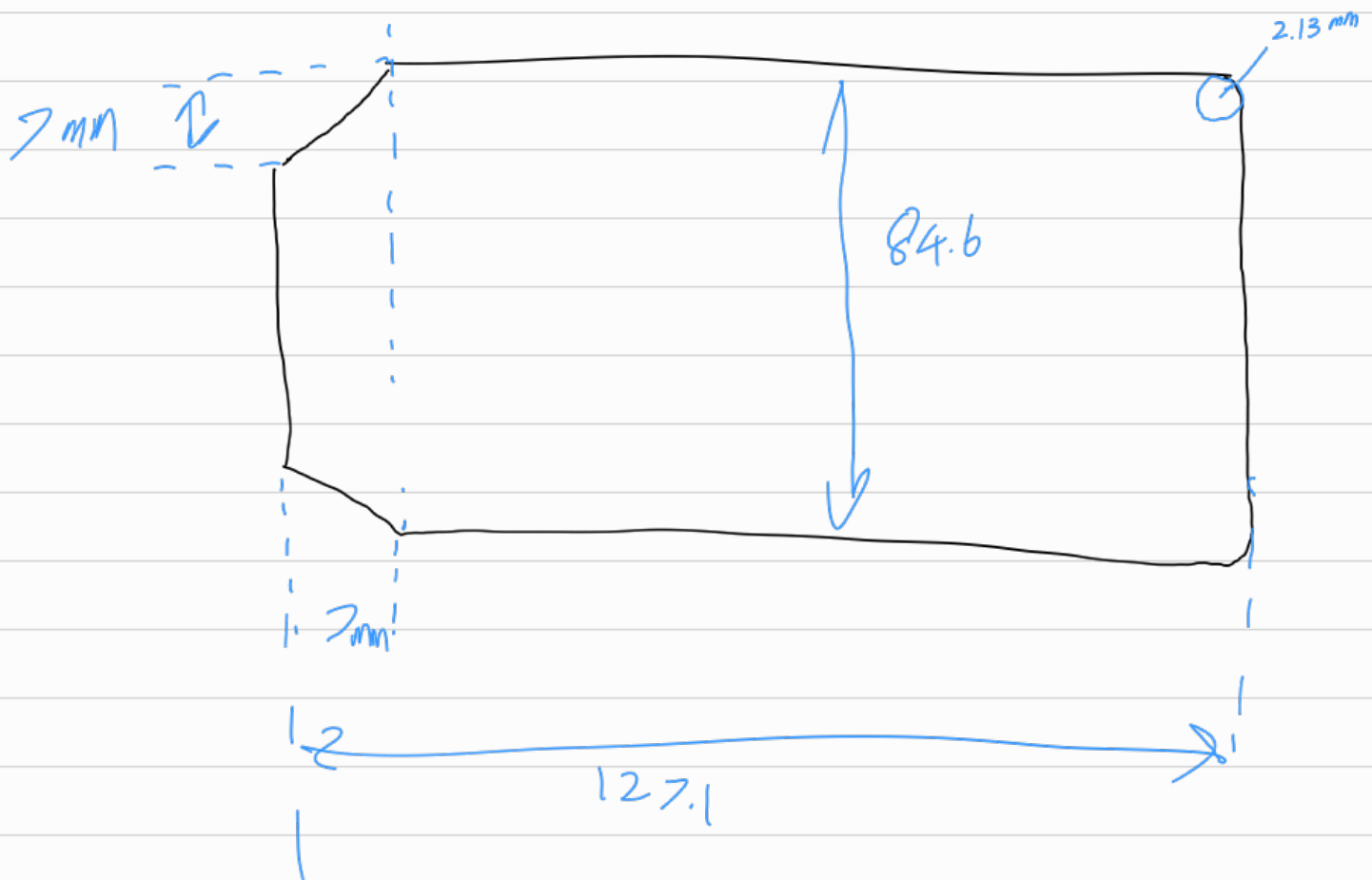
M.W Plate modeling:

Plate Model Found by Ethan Goldman on Onshape

Lid model made by mp

Lid

height 8 mm
cut in = 8 mil



Inner rib width: 1 mm

Length: 3.8 mm

height: 0.9 mm

Outer ridges : height 1 mm

; width 1.3 mm

